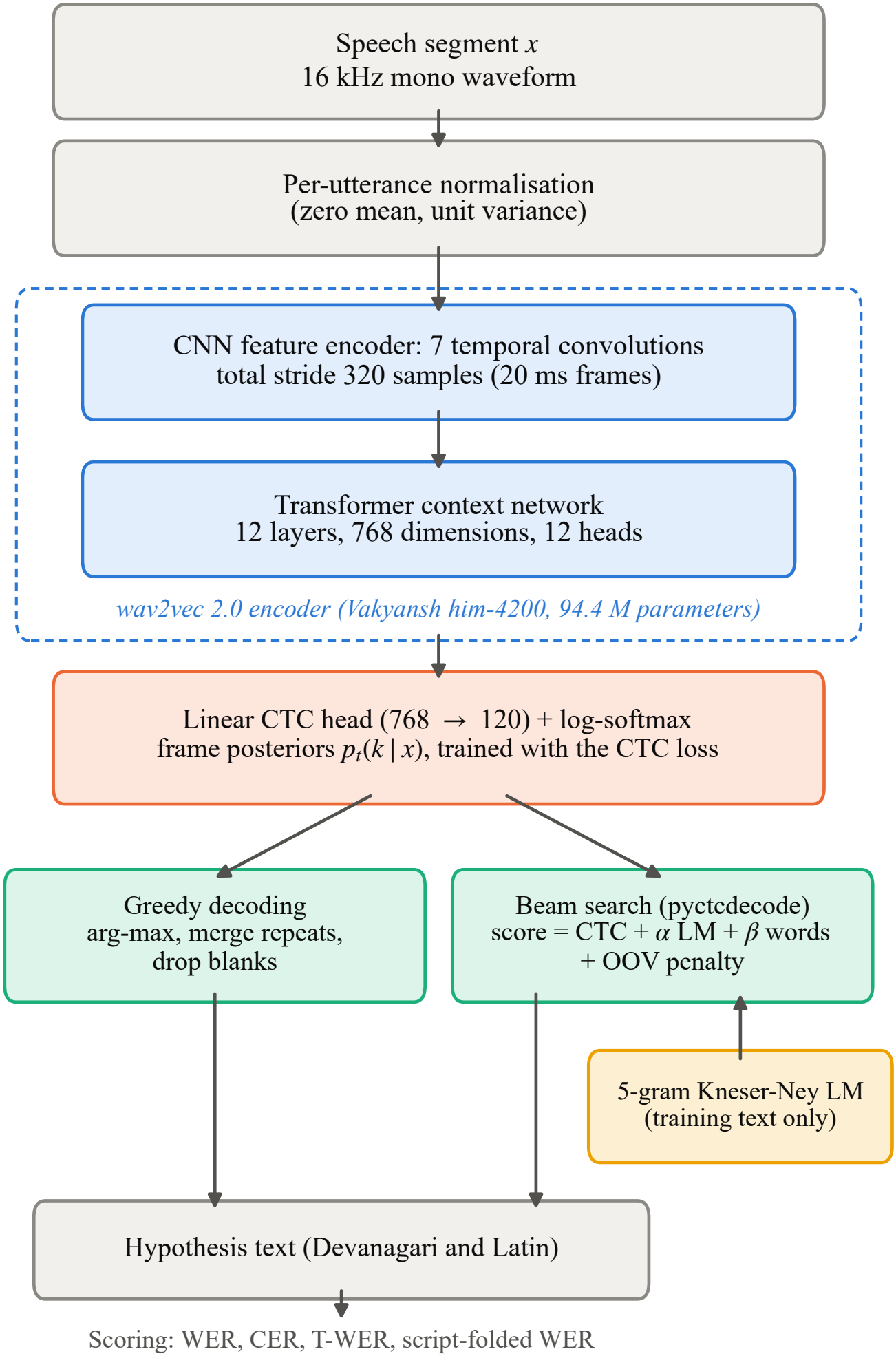

Speech segment x
16 kHz mono waveform
Per-utterance normalisation
(zero mean, unit variance)
CNN feature encoder: 7 temporal convolutions
total stride 320 samples (20 ms frames)
Transformer context network
12 layers, 768 dimensions, 12 heads
wav2vec 2.0 encoder (Vakyansh him-4200, 94.4 M parameters)
Linear CTC head (768 → 120) + log-softmax
frame posteriors $p_t(k \mid x)$, trained with the CTC loss
Greedy decoding
arg-max, merge repeats,
drop blanks
Beam search (pyctcdecode)
score = CTC + α LM + β words
+ OOV penalty
5-gram Kneser-Ney LM
(training text only)
Hypothesis text (Devanagari and Latin)
Scoring: WER, CER, T-WER, script-folded WER